QFun: Quantum Function Encoding with Signed and Quasi-Probability Extensions

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Abstract

Quantum amplitude encoding embeds a classical vector of $N = 2^n$ real numbers into the probability amplitudes of an n-qubit state, achieving an exponential compression of classical data into Hilbert space. A fundamental constraint is that measurement obeys the Born rule, so the observable distribution $p(i) = |\alpha_i|^2$ is always nonnegative. This paper presents *QFun*, an open-source simulation library that implements three layers of amplitude encoding: (i) standard nonnegative encoding for one-dimensional and d-dimensional functions on tensor-product grids, (ii) *Mode A*, an ancilla-based scheme that encodes the sign of each function value into an auxiliary qubit, and (iii) *Mode B*, a two-channel quasi-probability decomposition that splits a signed distribution into two proper distributions and recombines them classically. Mode B is motivated by the theory of negative probabilities developed by Dirac, Feynman, Bartlett, and recently revisited by Polson and Sokolov. We provide a rigorous mathematical framework for each mode, analyse statistical convergence, and demonstrate the library on the 27-equation Feynman benchmark dataset from the KAN paper.

1 Introduction

A central primitive in quantum computing is the ability to load classical data into a quantum register. Given a vector $\boldsymbol{v} = (v_0, \ldots, v_{N-1})^\top \in \mathbb{R}^N$ with $N = 2^n$, one defines the *amplitude-encoded* state

$$|\psi_{\boldsymbol{v}}\rangle = \frac{1}{\|\boldsymbol{v}\|_2} \sum_{i=0}^{N-1} v_i |i\rangle, \tag{1}$$

where $\{|i\rangle\}$ is the computational basis of n qubits. This encoding underlies quantum linear-system solvers [3], quantum machine learning kernels, and quantum Monte Carlo techniques.

The price of exponential compression is that direct measurement of $|\psi_{\boldsymbol{v}}\rangle$ in the computational basis yields outcomes distributed as $p(i) = |\alpha_i|^2 \propto v_i^2$, destroying all sign information. For nonnegative data this is acceptable: sampling from p recovers the *shape* of $\boldsymbol{v}$. For signed data the situation is more subtle and requires additional structure, either in the quantum circuit or in classical post-processing.

This paper describes **QFun**, a lightweight, CPU-only Python library built on PennyLane [2] that implements:

1. **Standard amplitude encoding** for nonnegative functions on 1D and d-dimensional grids (Sections 2.1–2.2).

2. **Mode A** — ancilla-based sign encoding (Section 2.4).

3. **Mode B** — two-channel quasi-probability decomposition (Section 2.5), grounded in the framework of Polson and Sokolov [5].
4. The **Feynman benchmark dataset** of 27 physics equations from the KAN paper [4], originally curated by AI Feynman [6] (Section 3).

2 Mathematical Framework

We develop the formalism in increasing generality: first for a single real variable, then for multivariate functions on tensor-product grids, and finally for signed targets.

2.1 Standard amplitude encoding in one dimension

Definition 1 (1D grid)**.** *Let $f : [a, b] \to \mathbb{R}_{\geq 0}$ be a nonnegative function and let $n \in \mathbb{N}$ be the number of qubits. Define $N = 2^n$ and the* uniform grid

$$x_i = a + i\,\Delta x, \quad i = 0, 1, \ldots, N-1, \qquad \Delta x = \frac{b-a}{N-1}. \tag{2}$$

Each grid index i has a unique n-bit binary representation and therefore corresponds one-to-one with the computational basis state $|i\rangle$ of the n-qubit Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$.

Definition 2 (Amplitude vector)**.** *The* raw amplitude *at grid point x_i is*

$$\tilde{\alpha}_i = \sqrt{f(x_i) + \varepsilon}, \tag{3}$$

where $\varepsilon > 0$ is a small regulariser (default 10^{-12}) that avoids numerical issues when $f(x_i) = 0$. The normalised amplitude vector *is*

$$\alpha_i = \frac{\tilde{\alpha}_i}{\|\tilde{\boldsymbol{\alpha}}\|_2}, \qquad \|\tilde{\boldsymbol{\alpha}}\|_2 = \left(\sum_{j=0}^{N-1} \tilde{\alpha}_j^2\right)^{1/2}, \tag{4}$$

so that $\sum_i |\alpha_i|^2 = 1$.

Proposition 1 (Sampling reproduces f)**.** *Let $|\psi\rangle = \sum_i \alpha_i |i\rangle$ be the state prepared by PennyLane's* `AmplitudeEmbedding`*. By the Born rule, a computational-basis measurement returns outcome i with probability*

$$p(i) = |\alpha_i|^2 = \frac{f(x_i) + \varepsilon}{\sum_{j=0}^{N-1} \big(f(x_j) + \varepsilon\big)} \xrightarrow{\varepsilon \to 0} \frac{f(x_i)}{\sum_j f(x_j)}. \tag{5}$$

Hence the empirical histogram of S independent shots converges to a distribution proportional to f evaluated on the grid.

Remark 1 (Statistical convergence)**.** *If $\hat{p}(i) = c_i/S$ is the relative frequency of outcome i in S shots (where c_i is the count for outcome i), then $\mathbb{E}[\hat{p}(i)] = p(i)$ and $\mathrm{Var}[\hat{p}(i)] = p(i)(1 - p(i))/S$. The expected total L_1 error satisfies $\mathbb{E}\big[\sum_i |\hat{p}(i) - p(i)|\big] = O\Big(\sqrt{N/S}\Big)$ by Cauchy–Schwarz and the multinomial variance identity.*

Remark 2 (Interpreting bitstrings)**.** *A measurement outcome such as* `001101` *is the binary representation of an integer index i. QFun converts $i \mapsto x_i$ via Eq.* (2) *so that the histogram can be plotted against the original domain $[a, b]$.*

Figure 1 illustrates the 1D encoding for three nonnegative functions: $f(x) = x^2$, $f(x) = e^{-x^2}$, and $f(x) = |x|$. In each panel the red curve is the target distribution $|\alpha|^2$ and the blue bars are the empirical histogram from 10^5 shots on a 6-qubit ($N = 64$) grid.

2.2 Extension to multivariate functions

Definition 3 (Tensor-product grid)**.** *For a d-variable function $f(x_1, \ldots, x_d)$, allocate n_k qubits to variable x_k with domain $[a_k, b_k]$, $k = 1, \ldots, d$. The 1D grids are*

$$x_k^{(j_k)} = a_k + j_k \, \Delta x_k, \quad j_k = 0, \ldots, 2^{n_k} - 1.$$

The tensor-product grid *is the Cartesian product of these 1D grids, containing*

$$N = 2^{n_1 + n_2 + \cdots + n_d} \tag{6}$$

points. QFun orders them in row-major (C-order) and stores them as a matrix $\boldsymbol{X} \in \mathbb{R}^{N \times d}$. The total qubit count is $n = n_1 + \cdots + n_d$.

Remark 3 (Asymmetric qubit budgets)**.** *Different variables may receive different qubit allocations. For example, one may assign $n_1 = 4$ qubits to a variable with fine structure and $n_2 = 3$ to a smoother variable, yielding a 16×8 grid on $n = 7$ qubits.*

Function values on the tensor-product grid are flattened to a length-N vector and amplitude-encoded exactly as in Section 2.1. The existing `run_shots` and `counts_to_distribution` routines therefore work without modification; only the grid construction and the mapping from flat index to multi-dimensional coordinates change.

Figure 2 shows a 2D example using Feynman equation I.12.11: $f(a, \theta) = 1 + a \sin \theta$.

2.3 The negativity problem

If $f(x_i) < 0$ for some grid points, the square-root mapping $\tilde{\alpha}_i = \sqrt{f(x_i)}$ produces imaginary values, and in any case $|\alpha_i|^2 = |f(x_i)|$ discards the sign. We therefore cannot directly recover signed targets from Born-rule measurements alone.

Historically, the idea that probabilities can be negative appears in Dirac's work on quantum electrodynamics (1942), Wigner's phase-space function (1932), and Feynman's discussion of negative probabilities (1987). Bartlett [1] formulated the key consistency condition: *negative probabilities must always be combined with positive probabilities to yield a proper (nonneg) probability distribution before any physical interpretation is admissible.* Polson and Sokolov [5] recently recast these ideas in a modern statistical framework of *extraordinary random variables.* QFun implements two complementary approaches motivated by this line of work.

2.4 Mode A: ancilla-based sign encoding

The first strategy attaches a single *ancilla qubit* to the data register to record the sign of each grid value.

Definition 4 (Signed amplitude state)**.** *Let f be a real-valued function on the grid and define the* sign mask

$$b_i = \begin{cases} 0 & \text{if } f(x_i) \geq 0, \\ 1 & \text{if } f(x_i) < 0. \end{cases} \tag{7}$$

The magnitude amplitudes *are $\alpha_i \propto \sqrt{|f(x_i)| + \varepsilon}$, normalised so that $\sum_i |\alpha_i|^2 = 1$. The* signed amplitude state *on $n+1$ qubits is*

$$|\Psi\rangle = \sum_{i=0}^{N-1} \alpha_i |i\rangle |b_i\rangle \;\in\; (\mathbb{C}^2)^{\otimes (n+1)}, \tag{8}$$

where the ancilla qubit $|b_i\rangle$ is the least-significant bit.

Proposition 2 (Signed reconstruction)**.** *Each computational-basis measurement of* $|\Psi\rangle$ *yields an* $(n+1)$*-bit string whose last bit is the sign bit* b*. Define the* conditioned empirical distributions

$$\hat{p}_+(i) = \frac{\#\{b = 0 \text{ at index } i\}}{S}, \qquad \hat{p}_-(i) = \frac{\#\{b = 1 \text{ at index } i\}}{S}. \tag{9}$$

Then $\hat{p}_+(i) + \hat{p}_-(i)$ *converges to a proper probability distribution (sums to 1), while the* signed quasi-probability

$$\hat{q}(i) = \hat{p}_+(i) - \hat{p}_-(i) \tag{10}$$

can take negative values and converges to $q(i) \propto \operatorname{sgn}(f(x_i))\,|f(x_i)|$ *as* $S \to \infty$*.*

Proof. By construction, for each index i the amplitude sits entirely at $|i\rangle|0\rangle$ or $|i\rangle|1\rangle$, never both. Therefore $p_+(i) = |\alpha_i|^2$ when $f(x_i) \geq 0$ and $p_+(i) = 0$ otherwise, with p_- complementary. Subtracting gives $q(i) = \operatorname{sgn}(f(x_i))\,|\alpha_i|^2 \propto f(x_i)$. □

Remark 4 (Circuit structure)**.** *The circuit uses* $n+1$ *wires. The first* n *wires carry the data register; the last wire is the ancilla. PennyLane's* `AmplitudeEmbedding` *prepares the* $2N$*-dimensional state vector of Eq.* (8) *in a single operation on the statevector simulator.*

2.5 Mode B: two-channel quasi-probability decomposition

The second strategy operates at the *distribution* level rather than the state level. It is directly motivated by the *extraordinary random variable* framework of Polson and Sokolov [5].

Definition 5 (Signed distribution and its decomposition)**.** *Let* $q : \{0, \ldots, N-1\} \to \mathbb{R}$ *be a signed distribution satisfying* $\sum_i q(i) = 1$ *with some entries negative. Define*

$$q^+(i) = \max\bigl(q(i),\, 0\bigr), \qquad q^-(i) = \max\bigl(-q(i),\, 0\bigr). \tag{11}$$

Let $Z_+ = \sum_i q^+(i) \geq 0$ *and* $Z_- = \sum_i q^-(i) \geq 0$ *(with* $Z_+ - Z_- = 1$*). The* normalised channels *are*

$$p_+(i) = \frac{q^+(i)}{Z_+}, \qquad p_-(i) = \frac{q^-(i)}{Z_-}, \tag{12}$$

which are proper (nonneg, normalised) probability distributions.

Proposition 3 (Exact reconstruction)**.** *The signed distribution is recovered by*

$$q(i) = Z_+\, p_+(i) \;-\; Z_-\, p_-(i). \tag{13}$$

Proof. $Z_+ p_+(i) - Z_- p_-(i) = q^+(i) - q^-(i) = q(i)$. □

Procedure. QFun amplitude-encodes p_+ and p_- independently:

1. Prepare $|\psi_+\rangle$ with amplitudes $\alpha_i^{(+)} \propto \sqrt{p_+(i)}$; sample S shots to obtain $\hat{p}_+$.
2. Prepare $|\psi_-\rangle$ with amplitudes $\alpha_i^{(-)} \propto \sqrt{p_-(i)}$; sample S shots to obtain $\hat{p}_-$.
3. Classically recombine: $\hat{q}(i) = Z_+ \hat{p}_+(i) - Z_- \hat{p}_-(i)$.

Proposition 4 (Signed expectations)**.** *For any observable* $g : \{0, \ldots, N-1\} \to \mathbb{R}$*,*

$$\mathbb{E}_q[g] = Z_+ \sum_i g(i)\, p_+(i) \;-\; Z_- \sum_i g(i)\, p_-(i) = Z_+\, \mathbb{E}_{p_+}[g] \;-\; Z_-\, \mathbb{E}_{p_-}[g]. \tag{14}$$

The estimator $\hat{\mathbb{E}}_q[g] = Z_+ \hat{\mathbb{E}}_{p_+}[g] - Z_- \hat{\mathbb{E}}_{p_-}[g]$ *is unbiased with variance*

$$\mathrm{Var}[\hat{\mathbb{E}}_q[g]] = \frac{Z_+^2\, \mathrm{Var}_{p_+}[g]}{S} + \frac{Z_-^2\, \mathrm{Var}_{p_-}[g]}{S}. \tag{15}$$

Remark 5 (Variance amplification)**.** *When the negative mass* Z_- *is large relative to* Z_+*, both channels contribute substantial variance and the effective sample size shrinks. This is the quasi-probability analogue of the* sign problem *in quantum Monte Carlo; it is an inherent cost of representing negativity via nonnegative sampling [5].*

Figure 3 demonstrates Mode A on Feynman equation I.50.26 ($f(a, o) = \cos a + o \cos^2 a$), which takes negative values. The diverging colourmap (red–blue) shows the target signed quasi-probability on the left and the measured reconstruction on the right.

3 Feynman Benchmark Dataset

To facilitate reproducible experiments on multivariate functions, QFun packages the 27-equation "Feynman" benchmark used in the KAN evaluation table [4] and originally curated for the AI Feynman symbolic-regression challenge [6].

Each equation is stored in its *dimensionless* form (fewer variables, no physical constants) as a `FeynmanEquation` named tuple with the following fields:

- `eq_id` — identifier (e.g. `I.12.11`),
- `formula` — human-readable formula string,
- `variables` — list of variable names,
- `domains` — default $[a_k, b_k]$ per variable,
- `func` — a NumPy-compatible callable.

The dataset spans equations with 1 to 6 input variables, covering Gaussians, trigonometric functions, power laws, and compositions thereof. Convenience functions `list_equations()` and `get_equation(eq_id)` provide programmatic access.

Table 1 lists all 27 equations with their dimensionless formulae, variable counts, and names.

Table 1: The 27 Feynman benchmark equations packaged in QFun (dimensionless forms).

ID	Dimensionless formula	#Vars	Variables
I.6.2	$\exp(-\theta^2/(2\sigma^2))/\sqrt{2\pi\sigma^2}$	2	$\theta,\ \sigma$
I.6.2b	$\exp(-(\theta-\theta_1)^2/(2\sigma^2))/\sqrt{2\pi\sigma^2}$	3	$\theta,\ \theta_1,\ \sigma$
I.9.18	$(a-c)^2 + (b-d)^2 + (e-f)^2$	6	a, b, c, d, e, f

continued on next page

ID	Dimensionless formula	#Vars	Variables
I.12.11	$1 + a \sin\theta$	2	$a,\ \theta$
I.13.12	$a(1/b - 1)$	2	$a,\ b$
I.15.3x	$a/\sqrt{1-b^2}$	2	$a,\ b$
I.16.6	$(a+b)/(1+ab)$	2	$a,\ b$
I.18.4	$(1+ab)/(1+a)$	2	$a,\ b$
I.26.2	$\arcsin(n \sin\theta_2)$	2	$n,\ \theta_2$
I.27.6	$1/(1/a + 1/b)$	2	$a,\ b$
I.29.16	$\sqrt{1 + a^2 - 2a\cos(\theta_1 - \theta_2)}$	3	$a,\ \theta_1,\ \theta_2$
I.30.3	$\sin^2(ab/2)/\sin^2(b/2)$	2	$a,\ b$
I.30.5	$\arcsin(a/b)$	2	$a,\ b$
I.37.4	$1 + a + 2\sqrt{a}\cos\delta$	2	$a,\ \delta$
I.40.1	$n_0 \exp(-a)$	2	$n_0,\ a$
I.44.4	$n \ln a$	2	$n,\ a$
I.50.26	$\cos a + o\cos^2 a$	2	$a,\ o$
II.2.42	$(a-1)b$	2	$a,\ b$
II.6.15a	$(a/c^2)\sqrt{b^2+c^2}$	3	$a,\ b,\ c$
II.11.7	$n_0(1 + p\cos\theta)$	3	$n_0,\ p,\ \theta$
II.11.27	$a + b$	2	$a,\ b$
II.35.18	$n_0/(\exp(a) + \exp(-a))$	2	$n_0,\ a$
II.36.38	$a + ab$	2	$a,\ b$
II.38.3	ab	2	$a,\ b$
III.9.52	$\sin^2((a-b)c/2)/((a-b)c/2)^2$	3	$a,\ b,\ c$
III.10.19	$\sqrt{1 + a^2 + b^2}$	2	$a,\ b$
III.17.37	$\beta(1 + \alpha\cos\theta)$	3	$\beta,\ \alpha,\ \theta$

4 Experiments

All experiments use PennyLane's `default.qubit` statevector simulator with NumPy as the backend (CPU only) [2]. Figures are reproducible via `paper/make_figures.py`.

4.1 One-dimensional examples

Three nonneg functions (x^2, e^{-x^2}, $|x|$) are encoded on 6-qubit grids and sampled with 10^5 shots. Figure 1 confirms that the empirical histograms closely match the target $|\alpha|^2$.

4.2 Two-dimensional Feynman examples

We encode Feynman I.12.11 ($1 + a\sin\theta$) on a 8×8 grid (3 qubits per variable, 6 total) and sample 10^5 shots. Figure 2 shows the 2D target vs. measured heatmaps.

4.3 Asymmetric qubit budgets

QFun supports non-uniform qubit allocation across variables. For example, encoding Feynman I.16.6 ($(a+b)/(1+ab)$) with 4 qubits for a and 3 for b produces a 16×8 grid on 7 total qubits, giving finer resolution along the variable of interest.

4.4 Signed multivariate example

Feynman I.50.26 ($\cos a + o \cos^2 a$) is negative in part of its domain. Mode A encodes the magnitudes on 6 data qubits plus one ancilla (7 total). Figure 3 shows the signed reconstruction from 10^5 shots.

5 Discussion and Limitations

Exponential grid growth. The grid size $N = 2^n$ grows exponentially in the number of qubits. For classical simulation this limits practical demonstrations to $n \lesssim 20$; on real quantum hardware the limit is set by gate fidelity and connectivity.

Discretisation error. The accuracy with which the sampled distribution represents the continuous function f depends on both the number of grid points (resolution) and the choice of domain $[a, b]$. Functions with sharp features require more qubits to resolve.

Shot noise. As noted in Remark 1, the expected L_1 reconstruction error scales as $O(\sqrt{N/S})$. For signed estimators (Mode B), the effective variance is amplified by a factor of $\|q\|_1^2 = (Z_+ + Z_-)^2$ relative to direct sampling (cf. Eq. (15)). Since $Z_+ - Z_- = 1$, this overhead equals $(1 + 2Z_-)^2$ and can be large when the negative mass Z_- is substantial.

State preparation cost. On real hardware, preparing an arbitrary amplitude vector requires $O(2^n)$ gates in general. Efficient preparation is an active research area; QFun uses PennyLane's built-in `AmplitudeEmbedding`, which is exact for statevector simulation.

6 Conclusion

We presented QFun, a reference implementation of quantum amplitude encoding for classical functions on 1D and multivariate grids. Beyond the standard nonneg setting, QFun provides two physically motivated extensions for signed targets:

- **Mode A** encodes the sign into an ancilla qubit, producing a signed quasi-probability via post-processing.
- **Mode B** decomposes a signed distribution into two proper distributions that are encoded and sampled independently, then recombined classically.

We also packaged the 27-equation Feynman benchmark for convenient multivariate demonstrations, and provided a complete mathematical framework — including convergence statements and variance analysis — for each encoding mode.

References

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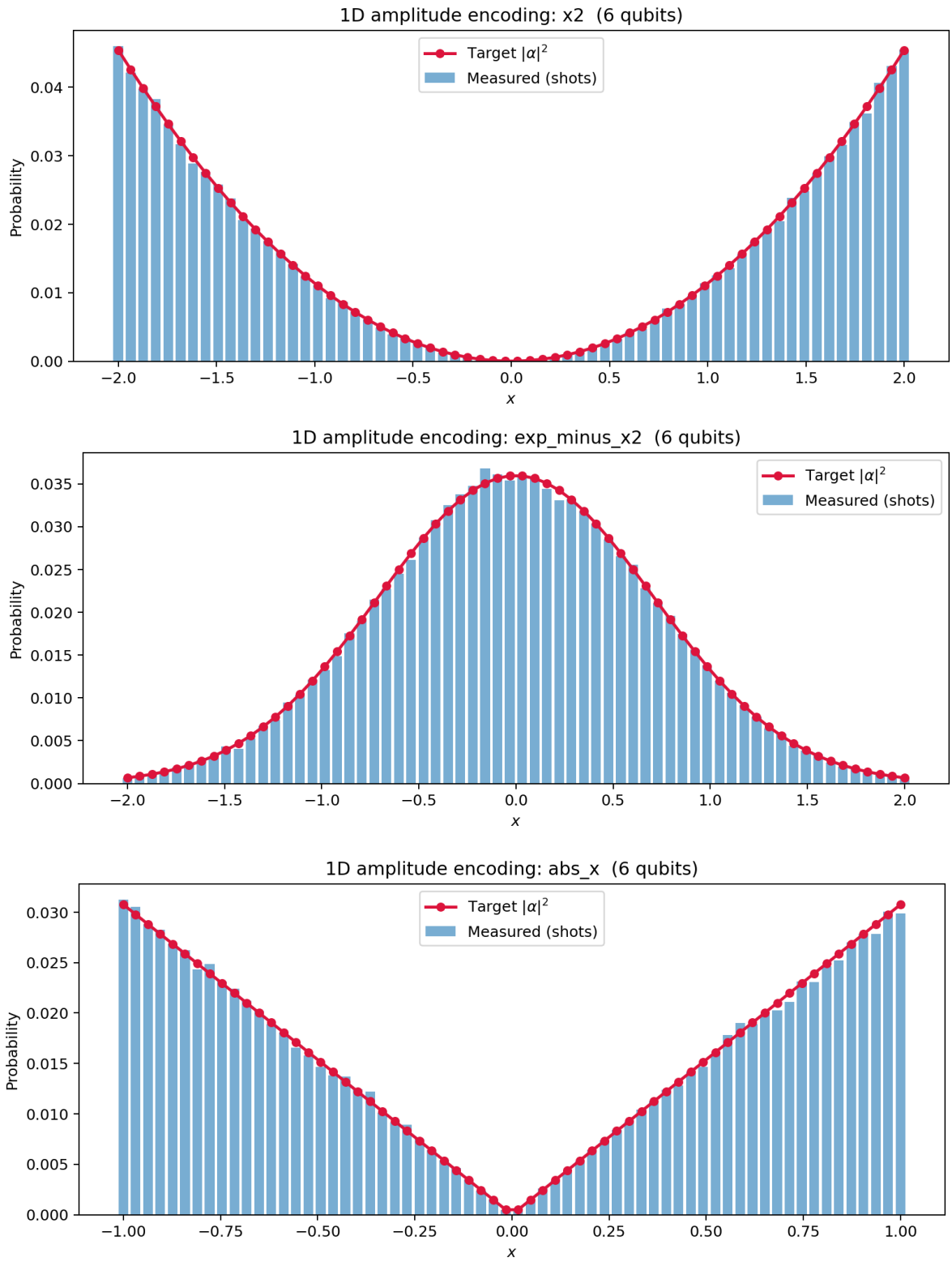

Figure 1: Standard 1D amplitude encoding for three nonnegative functions on a 6-qubit grid ($N = 64$ points, 10^5 shots). Red curve: target $|\alpha|^2$; blue bars: measured histogram.

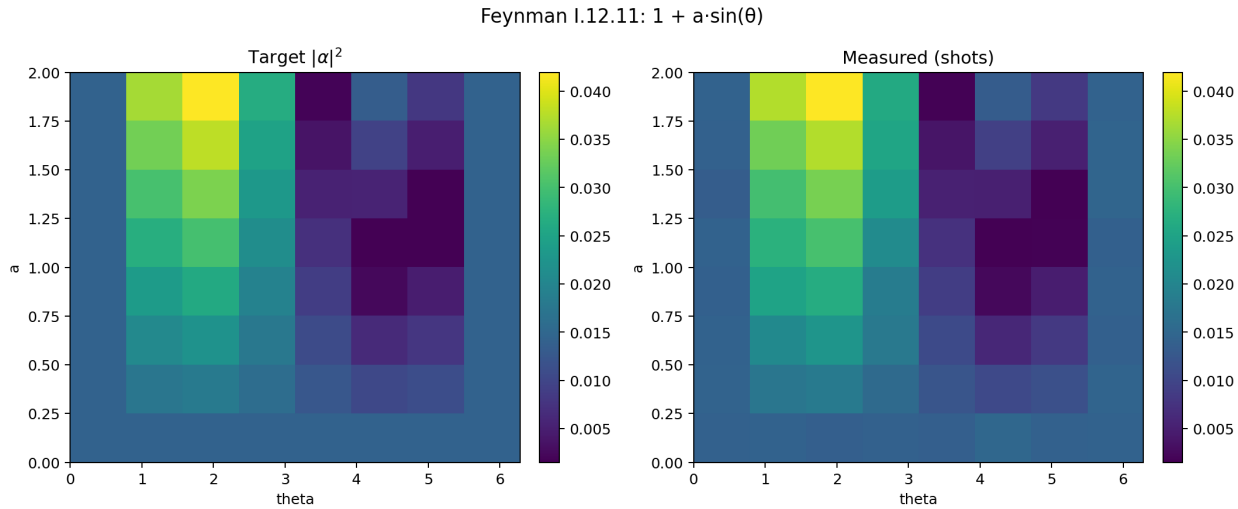

Figure 2: 2D amplitude encoding for Feynman I.12.11 ($f(a, \theta) = 1 + a \sin \theta$) on a 8×8 grid (6 qubits, 10^5 shots). Left: target $|\alpha|^2$. Right: empirical distribution.

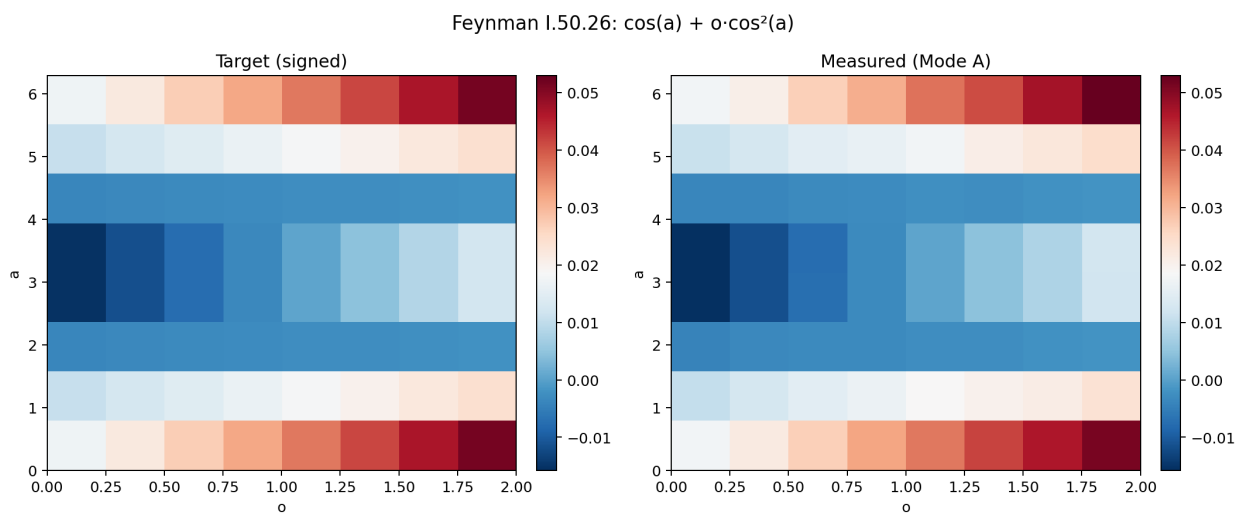

Figure 3: Signed (Mode A) 2D encoding for Feynman I.50.26. Left: target signed quasi-probability. Right: ancilla-based reconstruction from 10^5 shots.